

Engineering Economics (MG3008)

Date: February 23 2026

Course Instructor(s)

Mr. Syed Haris Mujtaba Kazmi

Sessional-I Exam

Total Time (Hrs): 1
Total Marks: 60
Total Questions: 3

Student Signature

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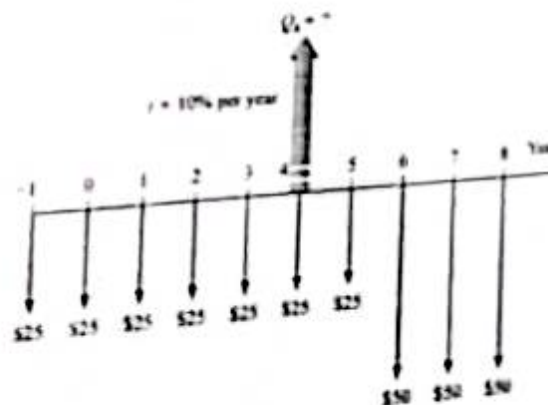
CLO # 1: Recognize the fundamental economic and financial considerations involved in engineering projects

Q1: Construct a table to repay the loan of \$180,000 in five years at 12 % per year compound interest such that equal payments are made each year including principal and accrued interest. [Marks: 25]

End of Year	Interest owed for year	Total owed at end of year	End of year payment	Total owed after payment
0				
1				
2				
3				
4				
5				

CLO # 1: Recognize the fundamental economic and financial considerations involved in engineering projects

Q2: For the cash flows shown in the diagram, compute the single amount of money Q_4 in year 4 that is equivalent to all of the cash flows shown. Use $i = 10\%$ per year. [Marks 25]



CLO # 1: Recognize the fundamental economic and financial considerations involved in engineering projects

Q3: If the value of Jane's retirement portfolio increased from \$170,000 to \$813,000 over a 15 year period, with no deposits made to the account over that period, calculate annual rate of return did she make.

[Marks 10]

Formula sheet

$\left(\frac{F}{P}, i, n\right) \Rightarrow F = P(1+i)^n$	$\left(\frac{P}{F}, i, n\right) \Rightarrow P = \frac{F}{(1+i)^n}$
$\left(\frac{P}{A}, i, n\right) \Rightarrow P = A \frac{(1+i)^n - 1}{i(1+i)^n}$	$\left(\frac{A}{P}, i, n\right) \Rightarrow A = P \frac{i(1+i)^n}{(1+i)^n - 1}$
$\left(\frac{F}{A}, i, n\right) \Rightarrow F = A \frac{(1+i)^n - 1}{i}$	$\left(\frac{A}{F}, i, n\right) \Rightarrow A = F \frac{i}{(1+i)^n - 1}$

Arithmetic Gradient $P = \frac{G}{i} \left[\frac{(1+i)^n - 1}{i} - \frac{n}{(1+i)^n} \right]$

Arithmetic Gradient $A = G \left[\frac{1}{i} - \frac{n}{(1+i)^{n-1}} \right]$

Geometric Gradient $P_g = A_1 \left\{ \frac{1 - \left(\frac{1+g}{1+i}\right)^n}{i-g} \right\} \quad g \neq i$

$P_g = A_1 \left(\frac{n}{1+i} \right) \quad g = i$

Effective $i = (1 + r/m)^m - 1$

Inflation Adjusted Interest Rate $i_f = i + f + if$

$AW_k = -P(A/P, i, k) + S_k(A/F, i, k) - \left[\sum_{j=1}^{j=k} AOC_j(P/F, i, j) \right] (A/P, i, k)$

Engineering Economics
(MG3008)

Date: April 6 2026

Course Instructor(s)

Mr. Syed Haris Mujtaba Kazmi

Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 75

Total Questions: 3

Roll No

Section

Student Signature

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CLO # 2: Interpret the concept of time value of money

Q1: Machines that have the following cost are under consideration for a robotized welding process. Using an interest rate of 10% per year, Determine which alternative should be selected on the basis of future worth analysis. [Marks: 25]

	Machine X	Machine Y
First cost, \$	-250,000	-430,000
Annual operating cost, \$/year	-60,000	-40,000
Salvage value, \$	70,000	95,000
Life, years	3	7

CLO # 2: Interpret the concept of time value of money

Q2: A total of \$100,000 was invested in two different projects identified as Z1 and Z2. If the overall rate of return on the \$100,000 was 30% and the rate of return on the \$30,000 invested in Z1 was 15%. Calculate the rate of return on Z2. [Marks 25]

CLO # 3: Analyze cost effectiveness of alternatives using engineering economy methods

Q3: Four methods are available to recover the lubricant from an automated milling system. The investment cost and income associated with each other are shown. Assuming that all methods have a 10-year life with zero salvage value, Compare the given methods and determine which one should be selected using a MARR of 10% per year and the incremental benefit cost method. Consider annual operating cost as a disbenefit. [Marks 25]

	1	2	3	4
First cost, \$	15,000	19,000	25,000	33,000
Annual operating cost, \$/year	10,000	12,000	9,000	11,000
Annual income, \$/year	16,000	20,000	19,000	22,000